

Business Rules

The rules that govern ordering, payment, the menu, inventory, staff and system administration at Bull's Coffee, a campus coffee shop.

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This document lists 96 business rules in 13 areas. Each rule has an ID (BR-01, BR-02, ...) so it can be referred to in requirements, test cases and reviews. The **Enforced by** column says who makes sure the rule is followed:

System: the Bull's Coffee software does it, or refuses anything that breaks it.

Policy: staff are responsible for it. The software doesn't check it, though it may display it or warn about it.

The last section lists rules the business needs that the software doesn't enforce yet.

1. Store operations

How the store runs day to day, and what the admin can switch on or off.

ID	Rule	Enforced by
BR-01	The store's posted hours are Monday to Friday 7:00 AM – 8:00 PM and Saturday 8:00 AM – 5:00 PM. The store is closed on Sundays.	Policy
BR-02	All prices and amounts are in Philippine pesos (₱), to two decimal places.	System
BR-03	All dates and times (orders, reports, schedules, backups) use store time: Philippine Time (UTC+8).	System
BR-04	The admin can pause kiosk ordering at any time. While it is paused, the kiosk refuses new orders and tells customers to order at the counter. Counter orders continue as normal.	System
BR-05	In maintenance mode, the kiosk stops taking orders and the website shows the admin's maintenance message (up to 500 characters). Turning maintenance mode on is flagged for admin review.	System
BR-06	The store name and support email set by the admin appear on every customer email.	System

2. Customers

Who counts as a customer, and how customer records are created and protected.

ID	Rule	Enforced by
BR-07	Customers sign in with an email and password or a social account. Their customer record is created automatically on first sign-in, using the account's name and email. If the account has no name, the customer is asked for one.	System
BR-08	Each customer has a unique email address. A university ID is optional, but no two customers can share one.	System

BR-09	Nobody has to sign in to buy. Guests order at the kiosk, and walk-ins are served at the counter without a customer record.	System
BR-10	A kiosk order is linked to a customer only when that customer is signed in on the device. A typed-in customer number is never accepted.	System
BR-11	At the register, the cashier may attach a customer to an order by searching for their name, university ID or email.	System
BR-12	Apart from the register's customer search, only admins can view, edit or delete customer records.	System
BR-13	A signed-in person with no staff record is a customer and cannot open any staff screen.	System

3. Menu and pricing

Products, categories, sizes and recipes.

ID	Rule	Enforced by
BR-14	Every product belongs to exactly one category. Category names and product names must each be unique.	System
BR-15	A product's price must be zero or more. The listed price is the Tall price.	System
BR-16	Drinks sold in sizes are priced as: Tall = listed price, Grande = listed price + ₱20, Venti = listed price + ₱40. A sized drink must be given a size; a product without sizes cannot be given one.	System
BR-17	A product marked unavailable (for example, sold out) cannot be ordered at the kiosk or rung up at the register.	System
BR-18	Kiosk prices always come from the current menu when the order is placed, never from the customer's screen. If the menu changed while the customer was ordering, they are asked to remove the item and add it again.	System
BR-19	A product that has ever been ordered cannot be deleted. It is marked unavailable instead, so past orders keep their details. A category cannot be deleted while it still has products.	System
BR-20	A product's recipe lists each ingredient at most once, with a quantity greater than zero. Deleting a product deletes its recipe.	System
BR-21	Every price change is recorded in the activity log with the old and new price.	System

4. Orders

What makes a valid order and how an order moves from placed to finished.

ID	Rule	Enforced by
BR-22	Every order has at least one item, and every item has a whole-number quantity of at least 1.	System
BR-23	A kiosk order can have at most 30 lines and at most 20 of any one item per line. Special instructions are limited to 200 characters per item.	System
BR-24	Each order records where it was placed: at the counter or at the kiosk.	System
BR-25	An order is Pending, Completed or Cancelled. Only a pending order can change status. Completed and cancelled orders are final.	System
BR-26	The order total is the subtotal minus any discount, and the system calculates it. The total can never be below zero, and neither the register nor the kiosk can set it.	System
BR-27	The cashier on a counter order is the employee signed in at the register. A kiosk order has no cashier until an employee takes its payment, completes it or cancels it. That employee then becomes its cashier.	System
BR-28	A paid order stays Pending while it is in the barista's queue. Staff mark it Completed when it is handed to the customer.	System
BR-29	An order can be completed only when it is paid in full.	System
BR-30	Staff with the cancel permission can cancel any pending order. Cancelling an order that has already been paid is flagged for admin review.	System
BR-31	Staff refund cancelled paid orders: online payments from the PayMongo dashboard, and others at the counter. The cancellation email asks the customer to reply so the refund can be arranged.	Policy
BR-32	To stop fake orders, one device or network can place at most 30 kiosk orders every 5 minutes.	System

5. Discounts

Who can get a discount, how much it can be, and where it is applied.

ID	Rule	Enforced by
BR-33	A discount is either a percentage or a fixed peso amount. Its value must be above zero, and a percentage cannot exceed 100%. Discount names are unique.	System
BR-34	Each discount has an eligibility rule: open to everyone, university ID required (students), or government ID required (for example, senior citizens or PWDs).	System
BR-35	A university-ID discount can be applied only when the order's customer has a university ID on record.	System
BR-36	Before applying a government-ID discount, the cashier must check the customer's ID at the counter.	Policy

BR-37	An order can carry at most one discount.	System
BR-38	Discounts are applied only at the register. Kiosk orders are charged the full menu price.	System
BR-39	The system calculates a preset discount's amount from its definition. The cashier may instead enter a custom amount. Either way, a discount can never exceed the order's subtotal.	System
BR-40	Only active discounts can be applied. A discount that has been used on any order cannot be deleted. It is switched off instead, so past orders keep its name.	System
BR-41	A discount that covers the entire subtotal (a free order) is flagged for admin review.	System

6. Payments

Accepted payment methods, and when an order counts as paid.

ID	Rule	Enforced by
BR-42	Accepted payment methods are cash, card and e-wallet. Online payments go through PayMongo and accept GCash, Maya, GrabPay, QR Ph and cards. GCash, Maya, GrabPay and QR Ph are recorded as e-wallet.	System
BR-43	Every payment must be for more than ₱0. An order whose total is ₱0 needs no payment.	System
BR-44	For cash, the register does not accept the payment until the cash received covers the total. The cashier gives change for the difference, and the receipt shows it.	System
BR-45	A kiosk order paid at the counter is charged its whole remaining balance in one payment. Only a pending order with a balance can be paid, and two cashiers cannot charge the same order at once.	System
BR-46	An online payment charges the order's outstanding balance. The order counts as paid only when PayMongo's signed confirmation arrives, never just because the customer reached the "payment successful" page.	System
BR-47	Each PayMongo payment is recorded once, even if PayMongo sends its confirmation more than once.	System
BR-48	If online checkout cannot start at the kiosk, the order still stands and the customer pays at the counter.	System
BR-49	If an online payment is confirmed for an order that was cancelled in the meantime, the payment is still recorded, because the money was taken. Staff must refund it.	System
BR-50	Online payment is offered only when the payment gateway is fully set up and at least one method is switched on. The admin chooses which methods are offered and whether PayMongo emails receipts.	System
BR-51	Card and e-wallet details are entered on PayMongo's own page and never pass through Bull's Coffee's systems.	System

7. Inventory

Ingredients, stock levels and how stock is allowed to change.

ID	Rule	Enforced by
BR-52	Each ingredient has a unique name and a unit of measure. Its minimum stock level must be zero or more.	System
BR-53	Stock can never fall below zero. Any change that would take it below zero is rejected.	System
BR-54	Stock levels change only through recorded stock movements (deliveries, sales, waste and adjustments) and are never edited directly. A new ingredient starts at zero, and its opening stock is entered as a movement.	System
BR-55	The only movements staff record by hand are waste and stock-count adjustments, and each must be a non-zero quantity.	System
BR-56	Every stock movement records when it happened and which employee made it. The employee is always the one signed in.	System
BR-57	An ingredient at or below its minimum stock level is shown to managers as Low Stock.	System
BR-58	Ingredients are retired by marking them inactive, not deleted.	System

8. Suppliers and deliveries

Who the store buys from, and how incoming stock is received.

ID	Rule	Enforced by
BR-59	Supplier names are unique, and each supplier needs a valid email address.	System
BR-60	A supplier's price list has one unit price per ingredient. Each price must be zero or more.	System
BR-61	A delivery comes from one supplier, has a delivery date, and is received by the employee who is signed in. It lists at least one ingredient. Each ingredient appears once, with a quantity above zero and a unit cost of zero or more.	System
BR-62	Recording a delivery adds every item to stock and logs a delivery movement for each one, all at once. If any part fails, none of it is recorded.	System
BR-63	Suppliers are retired by marking them inactive, not deleted.	System

9. Staff, roles and access

Who can do what, and the safeguards that keep the system from locking itself out.

ID	Rule	Enforced by
BR-64	Every staff member is a Cashier, Manager or Admin. New staff are cashiers unless set otherwise. Each staff email is unique.	System
BR-65	Staff join by invitation only. An invitation is claimed when the person signs in with the invited email address, and that address must be verified.	System

BR-66	Only active staff can use the staff screens. Deactivating a staff member also blocks them from signing in.	System
BR-67	What cashiers and managers can do is set by a permission matrix that admins edit. A change applies from the person's next action. The default grants are shown in the table after this section.	System
BR-68	Admins hold every permission. Seven system permissions (users and roles, activity logs, support tickets, system health, settings, payment gateway, and backups and exports) belong to admins only and cannot be granted to another role.	System
BR-69	Managers can add and edit cashiers and managers. Only an admin can make someone an admin.	System
BR-70	An admin cannot change their own role, or lock, deactivate or sign out their own account.	System
BR-71	The last active admin cannot be demoted or deactivated.	System
BR-72	Inviting a new admin, or promoting someone to admin, is flagged as a critical event for review.	System
BR-73	Staff records are retired by deactivating them, not deleted.	System

Default permissions

Admins can change the Cashier and Manager columns at any time (BR-67).

Permission	Cashier	Manager	Admin
View orders	Yes	Yes	Yes
Ring up orders and take payment	Yes	Yes	Yes
Cancel orders	Yes	Yes	Yes
View ingredients and stock levels	Yes	Yes	Yes
Manage menu items, prices and recipes	–	Yes	Yes
Manage discounts	–	Yes	Yes
Add ingredients, adjust stock and log waste	–	Yes	Yes
Manage suppliers and deliveries	–	Yes	Yes
Add and edit staff	–	Yes	Yes
View attendance	–	Yes	Yes
View sales reports	–	Yes	Yes
Export sales reports	–	Yes	Yes
Manage logs and PDF documents	–	Yes	Yes
<i>Seven system permissions (admin only)</i>	–	–	Yes

10. Scheduling and attendance

Weekly shifts, store coverage and time records.

ID	Rule	Enforced by
BR-74	Each staff member has one weekly shift: their working days plus a start and end time. A shift must include at least one working day, and its end time must differ from its start time. The default shift is Monday to Friday, 7:00 AM – 3:00 PM.	System
BR-75	Every day should be covered by at least 2 staff, with someone in by opening (7:00 AM) and someone until closing (9:00 PM). The schedule page warns about any day that falls short.	Policy
BR-76	A clock-out time must be after the clock-in time. If someone forgets to clock out, their open shift can be closed by entering the clock-out time.	System
BR-77	Attendance records can be viewed only by staff with the attendance permission (managers, by default).	System

11. Sales reporting

What counts as a sale and how reports are produced.

ID	Rule	Enforced by
BR-78	Only completed orders count toward sales. Pending and cancelled orders are left out.	System
BR-79	Gross sales are order totals before discounts. Net sales are gross sales minus discounts.	System
BR-80	Reports are daily (broken down by hour) or monthly (broken down by day), and list the top-selling products.	System
BR-81	Viewing reports and exporting them to CSV are separate permissions, both held by managers by default. An export made outside business hours (before 6:00 AM or after 10:00 PM) is flagged for review.	System

12. Customer communication and support

Order emails, and how complaints and bug reports are handled.

ID	Rule	Enforced by
BR-82	Customers are emailed when their order is placed, when it is ready, if it is cancelled, and when payment is received. Only orders linked to a customer get emails; walk-in and guest orders do not.	System
BR-83	Admins can edit each email template or switch it off. A failed or switched-off email never stops an order.	System
BR-84	Anyone can contact the store through the website's Contact form with a complaint or a bug report. They must give a subject (up to 150 characters), a message (up to 5,000), their name and a valid email address. Each message gets a ticket number (T-1, T-2, ...).	System
BR-85	One device or network can send at most 5 Contact messages per hour.	System

BR-86	A ticket is Open, In Progress, Resolved or Closed, and has a priority of Low, Medium, High or Urgent. New tickets start as Open with Medium priority.	System
BR-87	Every staff reply to a ticket is emailed to the person who reported it. Their replies go to the store's support inbox.	System

13. Audit, records and data protection

The audit trail, record keeping, backups and exports.

ID	Rule	Enforced by
BR-88	Sign-ins and changes are written to an activity log that cannot be edited or deleted. The only change allowed is marking a flagged entry as reviewed.	System
BR-89	These events are flagged and stay in the audit trail until an admin reviews them: a paid order cancelled; a full-value discount; a new admin invited or someone promoted to admin; maintenance mode turned on; a data export outside business hours; a database restore; a failed automatic backup.	System
BR-90	Records that past transactions depend on are retired rather than deleted: products are marked unavailable, discounts are switched off, and staff, ingredients and suppliers are deactivated.	System
BR-91	The database is backed up automatically on a schedule the admin sets: hourly, daily or weekly. Weekly backups run on Sundays, when the store is closed. The default is daily at 3:00 AM.	System
BR-92	Automatic backups are kept for 7, 30, 90 or 365 days (30 by default) and then deleted. Manual backups are kept until an admin deletes them. Only one backup runs at a time, and a backup that is still running cannot be deleted.	System
BR-93	Only an admin can restore the database, and they must type RESTORE to confirm. A restore either completes in full or changes nothing. It is always flagged as a critical event, and it never touches the activity log or the backup history.	System
BR-94	Admins can export orders, payments, customers, employees, inventory and the activity log as CSV or JSON. Export files are private and are deleted after 7 days.	System
BR-95	Backups, exports and uploaded CSV logs are stored privately. Only menu, supplier and staff images and PDF documents are public.	System
BR-96	The database, sign-in service, file storage, payment gateway and email are health-checked every minute. Results are kept for 30 days.	System

14. Rules not yet enforced

Gaps between what the business needs and what the software does today.

Sales don't use up ingredient stock yet. Recipes record how much of each ingredient a product uses, and "sale" is a stock movement type, but placing or completing an order doesn't deduct anything. Until it does, stock only goes down through waste and adjustments, so Low Stock warnings come too late.

Clock-in isn't recorded. Attendance rules cover clock-out, but nothing creates the clock-in record, so attendance has nothing to check against yet.

Counter prices aren't checked against the menu. The kiosk re-reads every price from the menu (BR-18). The register sends its own item prices, and the system accepts any price of zero or more.

Customer feedback can't be collected. The database has room for a 1–5 star rating and a comment per order, and managers have a Feedback page, but customers have no way to submit feedback yet.

Store hours don't match. The website shows Monday to Friday 7:00 AM – 8:00 PM, Saturday 8:00 AM – 5:00 PM, closed Sunday (BR-01). The staff schedule assumes 7:00 AM – 9:00 PM on every day, Sunday included (BR-75). One of the two needs to change.

Spam limits reset on restart. The kiosk-order and Contact-form limits (BR-32 and BR-85) are counted in the server's memory, so they reset whenever the server restarts.